

PetroQuant Oil Trading Strategy

An algorithmic trading system for crude oil (WTI) using machine learning and volatility modelling.
Built with Python | Updated: March 2026

What Does This Project Do?

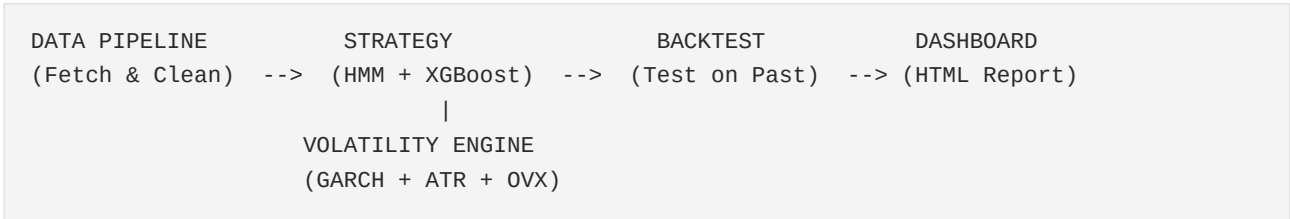
PetroQuant is an end-to-end algorithmic trading system that answers one question every trading day:

"Should I BUY, SELL, or HOLD crude oil today?"

It does this by:

- 1. **Fetching market data** Oil prices, volatility, inventories, dollar index, speculator positions, and more
- 2. **Detecting market conditions** Classifying the market into BULL, CHOPPY, or PANIC regimes
- 3. **Generating trade signals** BUY / SELL / HOLD with position sizing based on market conditions
- 4. **Forecasting future prices** Price range predictions from 1 day to 6 months ahead
- 5. **Backtesting** Validating the strategy on historical data with a detailed performance report
- 6. **Visualising everything** Interactive HTML dashboard with 10 panels

System Architecture



File Structure

File	Role
features.py	Defines which data sources to fetch
oil_data_pipeline_new.py	Fetches, cleans, and merges all market data
strategy.py	Core strategy HMM regime detection + XGBoost signals + Volatility Engine

File	Role
dashboard.py	Backtesting engine and interactive HTML dashboard
run_strategy.py	Entry point connects everything and runs the full pipeline

Supporting files:

- `.env` Stores API keys (`FRED_API_KEY`, `EIA_API_KEY`)
- `data/Rigcount_final.csv` Baker Hughes rig count (local CSV)
- `output/` Dashboard HTML and cached data files

Data Sources (9 Features)

Feature	Source	Update Frequency
WTI Crude Price	Yahoo Finance (<code>CL=F</code>)	Daily
Brent Crude Price	Yahoo Finance (<code>BZ=F</code>)	Daily
OVX (Oil Volatility Index)	FRED API	Daily
USD Index (DXY)	Yahoo Finance (<code>DX-Y.NYB</code>)	Daily
3-2-1 Crack Spread	Calculated from Yahoo Finance	Daily
Net Speculative Position	CFTC via Socrata API	Weekly
US Crude Inventories	EIA API	Weekly
Baker Hughes Rig Count	Local CSV	Monthly
Strategic Petroleum Reserve	EIA API	Weekly

All features are merged into a single `master_df` table with one row per trading day. Weekly/monthly data is forward-filled to maintain daily frequency.

Strategy Components

1. Feature Engineering

Raw prices are transformed into signals the model can learn from:

- **Log Returns** Daily and 5-day price changes for WTI and Brent
- **Z-Scores** How unusual the WTI-Brent spread, crack spread, and speculator positions are
- **Momentum** 10-day and 20-day price trends
- **Realised Volatility** 20-day rolling standard deviation of returns

- **Rate-of-Change** Changes in OVX, USD, inventories, rigs, and SPR over 15 days

Result: ~20 clean engineered features per trading day.

2. HMM Regime Detection

A **Hidden Markov Model (HMM)** classifies each trading day into one of three market regimes:

Regime	Behaviour	Avg Volatility
BULL	Calm, trending upward	Low (~18%)
CHOPPY	Sideways, uncertain	Medium (~25%)
PANIC	Large swings, risk-off	High (~42%)

The HMM is trained on daily returns and realised volatility to detect the hidden market state for each day.

Example output:

```
[+] BULL      +0.00080 (+20.2%/yr)  0.1800  950 days (51.9%)
[~] CHOPPY    +0.00010 (+2.5%/yr)  0.2500  468 days (25.6%)
[!] PANIC     -0.00100 (-25.2%/yr)  0.4200  412 days (22.5%)
```

3. Walk-Forward XGBoost Classifier

An **XGBoost** model predicts whether oil will be higher or lower 5 trading days from now.

Target variable:

```
1 (UP)   if WTI price in 5 days > today's price
0 (DOWN) if WTI price in 5 days < today's price
```

Walk-forward training is used to avoid data leakage the model is only ever tested on data it has never seen during training. It retrains every 63 trading days (~1 quarter).

Signal generation:

```
Probability > 55% --> BUY
Probability < 45% --> SELL
45% to 55%       --> HOLD (not enough conviction to trade)
```

Regime-aware position sizing scales down exposure in riskier markets:

```
BULL   regime = 1.00x position (full size)
CHOPPY regime = 0.50x position (half size)
PANIC  regime = 0.25x position (quarter size)
```

4. Volatility Engine Price Forecasting

The volatility engine generates **95% confidence intervals** for WTI prices across 7 time horizons (1d, 7d, 15d, 30d, 60d, 90d, 180d).

It uses a **composite of three volatility measures**:

- **GARCH(1,1)** Captures volatility clustering over medium horizons
- **ATR(14)** Average True Range, responsive to recent price action
- **OVX (Implied Vol)** The options market's forward-looking volatility estimate

Sample output:

Horizon	Price Range (95% CI)	Expected	Return
1d	\$56.85 -- \$59.07	\$57.97	+0.03%
7d	\$53.80 -- \$61.60	\$57.61	-0.58%
15d	\$51.13 -- \$67.10	\$59.12	+0.97%
30d	\$49.50 -- \$68.10	\$59.68	+1.97%

Backtesting Results

Backtested over ~2.9 years (732 trading days) of out-of-sample data.

Summary

Metric	Strategy	Buy & Hold
Total Return	+148.39%	-24.44%
Annualised Return	+36.78%	
Alpha (vs benchmark)	+172.84%	

Risk Metrics

Metric	Value
Sharpe Ratio	1.193
Sortino Ratio	1.687
Max Drawdown	-27.95%
Calmar Ratio	1.316
Win Rate	55.7%

Metric	Value
Up-Market Capture	35.2%
Down-Market Capture	16.3%

Dashboard 10-Panel Interactive Report

The dashboard is saved as `output/dashboard_hmm_xgb.html` and can be opened in any browser.

Panel	Contents
1	WTI price history with BULL/CHOPPY/PANIC regime shading
2	Buy/Sell signal overlay on price chart
3	Strategy equity curve vs buy-and-hold benchmark
4	Drawdown chart (peak-to-trough losses over time)
5	XGBoost feature importance
6	Model accuracy by market regime
7	Rolling Sharpe ratio over time
8	Monthly returns heatmap
9	Multi-horizon price forecast with 95% confidence bands
10	Full performance metrics table

How to Run

```
# 1. Add API keys to .env
FRED_API_KEY=your_key_here
EIA_API_KEY=your_key_here

# 2. Run the strategy
python run_strategy.py

# 3. Open the dashboard
output/dashboard_hmm_xgb.html
```

Dependencies: pandas, numpy, yfinance, arch, hmmlearn, xgboost, plotly, requests, python-dotenv

Key Terms

Term	Meaning
WTI	West Texas Intermediate US crude oil benchmark price
Regime	Market condition: BULL (calm), CHOPPY (uncertain), PANIC (volatile)
HMM	Hidden Markov Model detects hidden market states
XGBoost	Gradient boosting algorithm used for price direction prediction
GARCH	Model for estimating time-varying volatility
ATR	Average True Range measures typical daily price movement
OVX	Oil Volatility Index options market's fear gauge for crude oil
Walk-Forward	Training method where the model is always tested on unseen future data
Sharpe Ratio	Return divided by volatility measures risk-adjusted performance
Max Drawdown	Largest peak-to-trough loss during the backtest period
Alpha	Strategy return minus the benchmark (buy-and-hold) return
Crack Spread	Refinery profit margin from crude oil to refined products
COT / NSP	CFTC Commitment of Traders net speculator positioning in oil futures